

# CARECHRONICLE

## Design Approach

UX, Visual System, Product Architecture & Design Diagrams

**Visual direction:** Premium, calm, trustworthy — with a distinctive pink-led identity.

| Document         | Design Approach v2.0                                  |
|------------------|-------------------------------------------------------|
| Product          | CareChronicle                                         |
| Scope            | Product design + UX + UI + architecture visualization |
| Primary use case | Patient-owned medical record & care organization      |
| Initial focus    | Cancer / complex longitudinal care                    |
| Design palette   | Pink-led, high-contrast, restrained accent usage      |

*Organize the records. Preserve the story. Simplify the journey.*

# 1. Design North Star

CareChronicle should feel like a calm, trustworthy medical filing system — not a flashy chatbot.

The interface should transform a messy collection of reports into a coherent journey without pretending that the system is a clinician. The product earns trust through preservation, traceability, predictable organization and controlled AI assistance.

**North-star experience:** *“I can open CareChronicle and quickly understand where my records are, what happened recently, what changed, and what I should take to the next appointment.”*

Design success is measured by reduced searching, reduced reconstruction of history, faster appointment preparation and confidence that the original evidence remains available.

## 2. Core Design Principles

| Principle                  | Design approach                                                                    |
|----------------------------|------------------------------------------------------------------------------------|
| Trust before automation    | AI suggests; people verify important relationships and metadata.                   |
| Original evidence first    | Every derived insight can lead back to the original document.                      |
| Timeline over file pile    | The journey is the mental model; files support each event.                         |
| Appointment as a core unit | A visit groups reports, prescriptions, notes, bills and instructions.              |
| Show uncertainty           | Use explicit review states instead of silent assumptions.                          |
| Calm visual hierarchy      | Use whitespace and progressive disclosure; avoid dashboard overload.               |
| AI is the engine           | AI appears inside useful workflows rather than becoming the product identity.      |
| Patient control            | Sharing, access, deletion and export are explicit and understandable.              |
| Messy reality is normal    | Design for scans, photos, duplicates, conflicting dates and incomplete extraction. |

## 3. Visual Design Direction — Pink System

Pink is the primary brand and interaction color. It should communicate warmth and confidence without making the interface feel playful or cosmetic. The safest approach is a **deep pink primary + pale pink surfaces + neutral backgrounds + restrained semantic colors**.

### Core palette

| Token           | Suggested value | Usage                                                  |
|-----------------|-----------------|--------------------------------------------------------|
| Primary Pink    | #C2185B         | Primary actions, active navigation, key brand elements |
| Dark Pink       | #8E1748         | Headings, strong emphasis, cover/title treatment       |
| Accent Pink     | #E91E63         | Secondary interactive emphasis, selected states        |
| Light Pink      | #FCE4EC         | Callouts, selected surfaces, subtle backgrounds        |
| Soft Pink       | #F8BBD0         | Borders, dividers, low-emphasis highlights             |
| Neutral Surface | #FFF7FA         | Very light tinted app surfaces                         |
| Text            | #26202A         | Primary text                                           |
| Muted Text      | #6B6570         | Secondary metadata                                     |
| Success         | #287A57         | Confirmed/complete states only                         |

| Token   | Suggested value | Usage                                 |
|---------|-----------------|---------------------------------------|
| Warning | #9A6700         | Needs review/uncertainty              |
| Error   | #A33A3A         | Actual errors or destructive warnings |

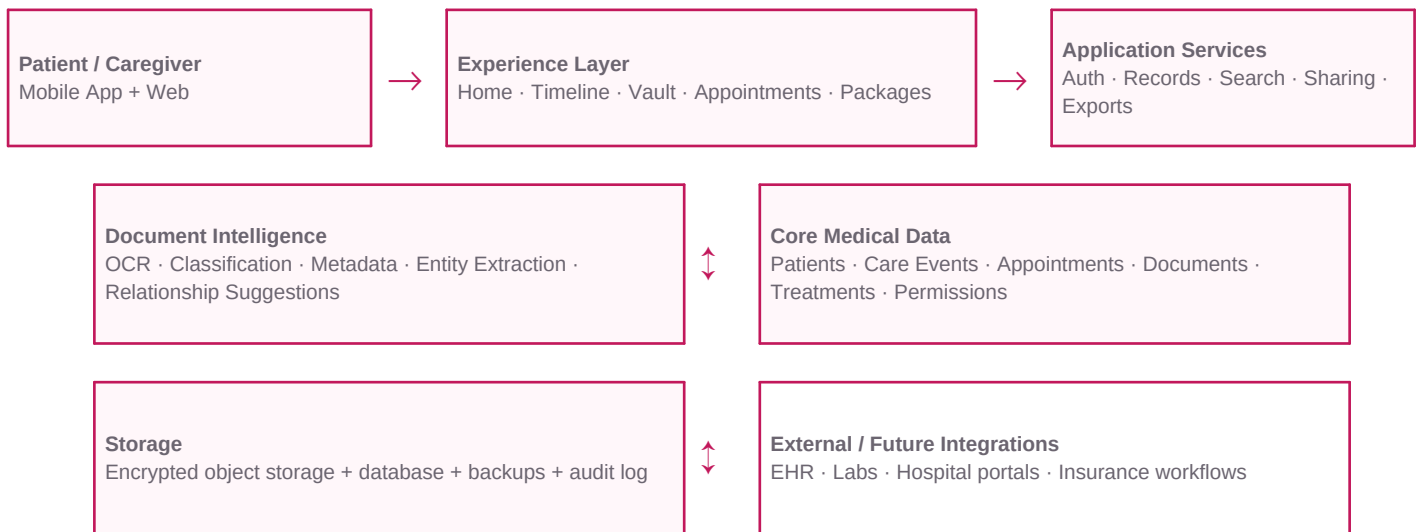
**Important:** semantic colors must not replace text labels. Pink is the product identity; green, amber and red remain functional status colors.

## 4. Information Architecture

| Area             | Purpose                                       | Primary actions                   |
|------------------|-----------------------------------------------|-----------------------------------|
| Home             | Current context and next action               | Add, review, prepare, search      |
| Timeline         | Longitudinal care story                       | Explore events, open evidence     |
| Medical Vault    | Complete document repository                  | Search, filter, preview, link     |
| Appointments     | Visit-centered organization                   | Prepare, review, add, export      |
| Packages         | Purpose-built outputs                         | Doctor, second opinion, insurance |
| Profile & Access | Identity, caregivers, permissions and privacy | Manage access, sharing, controls  |

## 5. System Architecture — Design View

The architecture should reinforce the product philosophy: preserve originals, derive structured information, connect it to care events, and present useful workflows through a patient-facing experience.



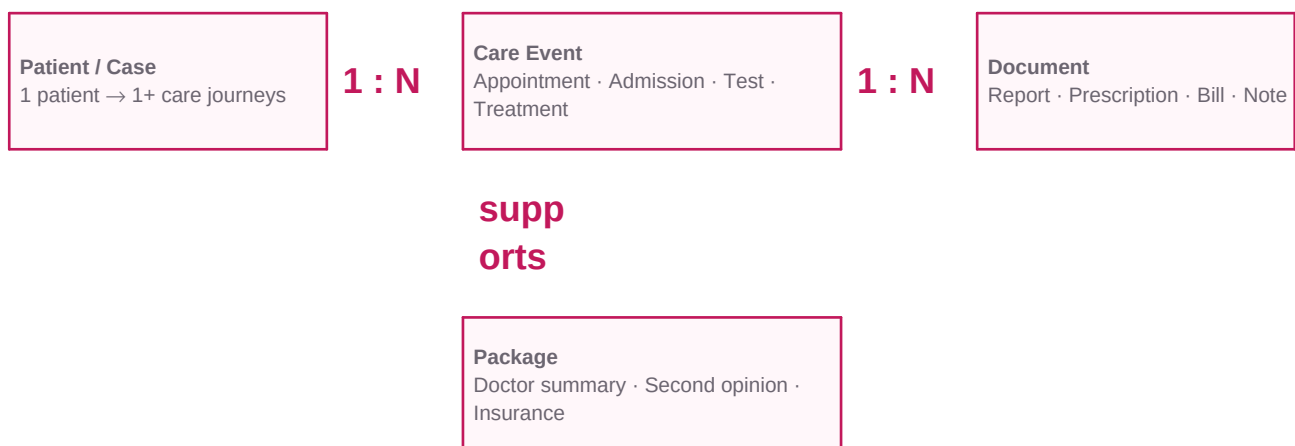
**Architecture principle:** the AI/document-intelligence layer should never become the system of record. The source document and verified medical-record entities remain the authoritative foundation.

### Design implications

- The UI can show AI processing as a supporting state, but the user's records remain accessible even if AI processing fails.
- Core navigation should work from structured records and documents; it should not require a chat session.
- Future integrations should feed the same care-event/document model rather than create separate silos.
- Exports should be generated from structured data + selected originals, with a manifest of included documents.

## 6. Core Data Relationship Diagram

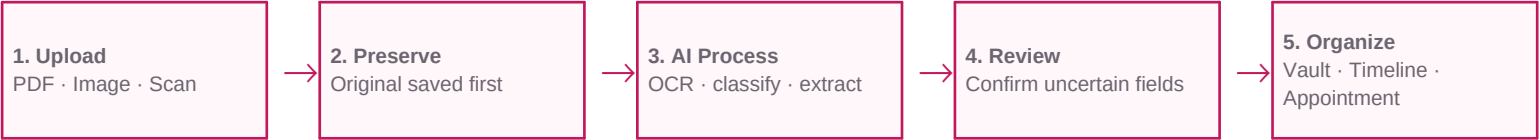
The design system should be based on explicit relationships rather than folders alone. This allows the same document to be discoverable through the Vault, appointment, timeline and package workflows.



| Object         | Design meaning                                                        |
|----------------|-----------------------------------------------------------------------|
| Patient / Case | The ownership and context boundary for the medical journey.           |
| Care Event     | An appointment, admission, test, treatment or other meaningful event. |
| Document       | The preserved source artifact and its metadata.                       |
| Package        | A curated output assembled for a specific purpose.                    |

# 7. Upload & Document Intelligence Flow

Upload is the first major trust interaction. The user should not need to understand OCR, classification or metadata schemas.



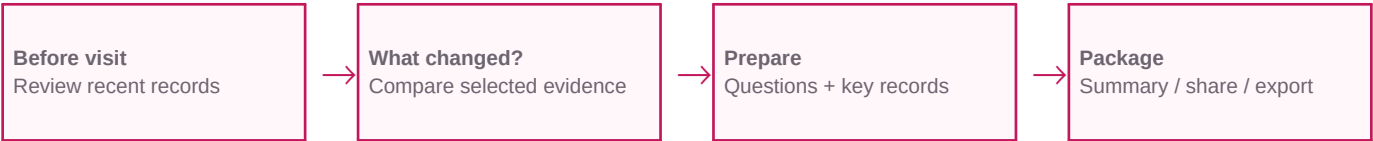
## Recommended interaction rule

- **Save first, enrich second:** the original is preserved before the system asks for review.
- **Ask only when it matters:** low-confidence or consequential associations should be confirmed.
- **Never block on perfection:** an unlinked document can remain safely in the Vault.
- **Make correction easy:** wrong classification or date should be fixable in seconds.

Metadata should distinguish at least document date, appointment date and billing date. These dates should not be forced into a single universal timestamp.

# 8. Appointment-Centered Experience

Appointments are the bridge between raw records and real-world care. A patient should be able to open one visit and see the evidence and actions surrounding it.



| Appointment area  | Design content                                                                 |
|-------------------|--------------------------------------------------------------------------------|
| Header            | Date, provider/specialty, status and context.                                  |
| Since last visit  | New records, changes, new prescriptions and relevant events.                   |
| What changed?     | Evidence-based comparison with source links.                                   |
| Questions / notes | Patient-entered notes and questions, clearly separated from clinician records. |
| Documents         | Reports, prescriptions, notes, bills, receipts and instructions.               |
| Actions           | Generate summary, curate package, share/export.                                |

# 9. “What Changed?” UX Architecture

This is a signature workflow. It should present recorded differences rather than diagnose or interpret them.

| Step                          | UI behavior                                                              |
|-------------------------------|--------------------------------------------------------------------------|
| 1. Choose comparison          | Select latest vs previous report/event or use the suggested pair.        |
| 2. Show extracted differences | Previous value → latest value, document section and date where reliable. |
| 3. Attach provenance          | Every meaningful difference links to the source page/document.           |
| 4. Flag uncertainty           | If extraction is incomplete or ambiguous, say so clearly.                |

| Step                          | UI behavior                                                           |
|-------------------------------|-----------------------------------------------------------------------|
| 5. Optional generated summary | Plain-language synthesis labeled as AI-generated and evidence-linked. |

**Safety boundary:** the interface can say what records show and how records differ. It should not convert those differences into diagnosis, prognosis or treatment advice.

## 10. Doctor / Appointment Preparation Design

The output should be concise enough for a real consultation while retaining access to the full record.

- **Since last visit:** the most relevant new events and records.
- **Key records:** curated evidence with one-tap originals.
- **Questions:** patient-entered items.
- **Summary:** structured one-page output.
- **Share/export:** explicit package contents before sending.

*Generated summaries should be evidence-linked. The design should make it possible to move from a summary statement to the underlying source record.*

## 11. Action Packages — Second Opinion & Insurance

| Package        | Design objective                                               | Core interaction                                                         |
|----------------|----------------------------------------------------------------|--------------------------------------------------------------------------|
| Second opinion | Create a professional, curated clinical record packet.         | Select care period → review records → preview → export/share             |
| Insurance      | Organize potentially relevant medical/financial documentation. | Select episode → gather docs → review what is present/not found → export |
| Doctor summary | Give a clinician a concise longitudinal context.               | Select visit → review changes → generate → verify → share                |

Insurance UX must never imply that CareChronicle knows an insurer's exact requirements unless those requirements have been explicitly supplied/verified. Prefer language such as “not found in CareChronicle” rather than “missing required document.”

## 12. AI Trust & Provenance System

| Information state | Meaning                                 | UI treatment                  |
|-------------------|-----------------------------------------|-------------------------------|
| Original          | Direct source artifact                  | Source document action        |
| Extracted         | Machine-read value                      | Extracted label + source      |
| User-entered      | Added or corrected by user              | Added by you                  |
| Suggested         | AI-inferred classification/relationship | Suggested + review action     |
| Generated         | AI-created summary text                 | AI-generated + evidence links |

**Visual hierarchy rule:** generated text must never look more authoritative than the source document.

## 13. Edge-Case & Failure-State Design

| Situation | Design response                                                                |
|-----------|--------------------------------------------------------------------------------|
| Poor scan | Preserve original; show extraction quality issue; allow retry/manual metadata. |

| Situation               | Design response                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| Handwriting             | Do not imply reliable extraction; keep original and allow manual classification. |
| Duplicate               | Suggest possible duplicate; user decides whether to keep.                        |
| Conflicting dates       | Show document/appointment/billing dates separately.                              |
| No appointment match    | Keep in Vault; allow later linking.                                              |
| Wrong AI classification | Fast correction; corrected value becomes the trusted user-confirmed value.       |
| Incomplete extraction   | Expose uncertainty instead of filling gaps silently.                             |
| Multiple cases          | Explicit case context before association to prevent cross-case filing.           |



## 14. Screen-by-Screen Design Approach

| Screen                | Primary job                           | Key elements                                               |
|-----------------------|---------------------------------------|------------------------------------------------------------|
| Onboarding            | Establish ownership/context           | Patient/case setup, privacy explanation, first upload      |
| Home                  | Orient user quickly                   | Next appointment, recent care, review queue, quick actions |
| Upload                | Add evidence with low friction        | Picker/camera, progress, original preservation             |
| AI Review             | Confirm important machine suggestions | Suggested type/date/provider/appointment + edit            |
| Vault                 | Retrieve any record                   | Search, filters, document cards, relationship context      |
| Document Viewer       | Inspect source evidence               | Full document, metadata, linked event, share               |
| Timeline              | Understand the journey                | Care events, milestones, supporting documents              |
| Appointments          | Prepare around visits                 | Visit list, preparation state, linked records              |
| Appointment Workspace | Manage one care event                 | Since-last-visit, documents, notes, package actions        |
| Packages              | Create useful outputs                 | Doctor, second opinion, insurance package builder          |
| Access & Privacy      | Maintain control                      | Caregivers, permissions, sharing history, data controls    |

## 15. Design System Components & States

| Component        | Required states                                                 |
|------------------|-----------------------------------------------------------------|
| Document card    | Default · Processing · Needs review · Verified · Error · Shared |
| Appointment card | Upcoming · Completed · Incomplete · Archived                    |
| Timeline event   | Collapsed · Expanded · Selected · Missing evidence              |
| AI suggestion    | Suggested · Accepted · Rejected · Edited                        |
| Upload           | Idle · Uploading · Processing · Success · Partial failure       |
| Package builder  | Selected · Omitted · Uncertain · Ready to export                |
| Permission       | View · Contribute · Export/share · Revoked                      |

## 16. Mobile, Web & Print Strategy

| Context     | Design priority                                                                            |
|-------------|--------------------------------------------------------------------------------------------|
| Mobile      | Upload, appointment prep, search, timeline, quick access, sharing.                         |
| Desktop/Web | Bulk document review, side-by-side source inspection, package building and administration. |
| PDF/Print   | Professional summaries/packages with clear dates, provenance and document manifest.        |

Desktop should not simply be a stretched mobile UI. Use larger space for document preview, metadata review and package curation.

## 17. Accessibility & Localization

- Target WCAG 2.2 AA as the design baseline where applicable.
- Never communicate status through color alone.
- Maintain comfortable touch targets and scalable text.

- Use plain-language action labels.
- Support screen readers with meaningful document/status labels.
- Design for older users and users under stress.
- Support multilingual content inside documents without destroying the original.
- If translation is offered, label it as translation/generated content and preserve the source.

## 18. Privacy & Sharing UX

Privacy is part of the primary experience. A patient should always know what is being shared, with whom and for what scope.

- Show current access to a case.
- Make package contents visible before sharing.
- Use explicit recipient and scope confirmation.
- Support revocation/expiry where implemented.
- Show sharing history.
- Make deletion/export controls discoverable.
- Explain AI processing in plain language.
- Avoid public-by-default sharing.

The key trust question is: **“Did I just share my entire medical history?”** The interface must make the answer unambiguous.

## 19. Design Validation & Testing

| Test             | Question                                    | Success signal                                     |
|------------------|---------------------------------------------|----------------------------------------------------|
| Card sort        | Do users understand the organization model? | Strong agreement on core categories                |
| Upload prototype | Do users trust AI suggestions?              | Users can verify/correct without confusion         |
| Retrieval test   | Can users find a requested record quickly?  | Important record found without manual file hunting |
| Appointment test | Does the appointment model make sense?      | User prepares a visit without instruction          |
| Comparison test  | Is “What Changed?” understandable and safe? | Users identify recorded differences and sources    |
| Sharing test     | Is package scope clear?                     | No ambiguity about recipient/content               |

## 20. Design Metrics

| Metric                             | Definition                                                |
|------------------------------------|-----------------------------------------------------------|
| Time to organized record           | Upload to useful placement in Vault/timeline/appointment. |
| Retrieval time                     | Time to locate a requested record.                        |
| AI correction rate                 | How often suggestions require user correction.            |
| Appointment preparation completion | Eligible visits for which preparation is completed.       |
| Source traceability                | Generated summary items traceable to source evidence.     |
| Sharing comprehension              | Users understand what a package contains before sharing.  |

| Metric      | Definition                                        |
|-------------|---------------------------------------------------|
| Trust score | User confidence in preservation and organization. |

## 21. Final Design Direction

CareChronicle should not try to look like the smartest healthcare app. It should feel like the **most dependable place to organize and retrieve a patient's medical story.**

The pink visual identity should make the product distinctive while the underlying UX remains restrained, professional and evidence-first.

Every major interaction should reinforce three ideas:

- **Your records are preserved.**
- **Your journey is organized.**
- **You remain in control.**

The strongest product moment is not “Ask AI about my report.” It is: **“Everything related to my care is here, I can see what happened, I can see what changed, and I can take the right records to the next person who needs them.”**

*Design mantra: **Organize the records. Preserve the story. Simplify the journey.***

This document should be used with the CareChronicle PRD as the UX/UI and architecture-visualization foundation for the design and engineering teams.